

## Problem 1.

Let's think about  $B \rightarrow B'$ ,  $19.4 = \frac{1}{4}(5 + r \cdot 16.0) \times 4 = 5 + 16r$

$$\therefore 16r = 14.4 \quad r = \frac{144}{160} = \frac{9}{10} = 0.9$$

$$\begin{aligned} \text{Let the } V_{\pi}(A) &= x, \quad x = \frac{1}{4}(10 + r^5 \left( \frac{1}{4}(10 + r^5(\dots)) \times 4 \right)) \times 4 \\ &= 10 + r^5(10 + r^5(10 + r^5 \dots \\ &= 10 + r^5 \cdot 10 + r^{10} \cdot 10 + \dots \\ &= \sum_{n=0}^{\infty} 10 r^{5n} = \frac{10}{1-r^5} \quad (r < 1, \therefore r^5 < 1) \end{aligned}$$

Let  $r = 0.9$ ,

$$\therefore x = 24.41942809 \dots$$

$$\therefore x = 24.419$$

$$\text{Ans: } x = 24.419$$

## Problem 2.

1)  $V_{\pi}(s) = E[G_t | S_t = s]$ , by  $G_t = r + \gamma V_{\pi}(s')$  (which  $s'$  is next state of  $s$ )

$$\begin{aligned} \therefore V_{\pi}(s) &= E[r + \gamma V_{\pi}(s') | S_t = s] \\ &= \sum_a E[r + \gamma V_{\pi}(s') | S_t = s, A_t = a] \cdot P(A_t = a | S_t = s) \\ &= \sum_a \pi(a|s) E[r + \gamma V_{\pi}(s') | S_t = s, A_t = a] \\ &= \sum_a \pi(a|s) \sum_{s', r} P(s', r | s, a) [r + \gamma V_{\pi}(s')] \quad \dots \textcircled{1} \\ &\quad (\because r, s' \text{ is both random variable, } G_t = V_{\pi}(s')) \end{aligned}$$

2) reward is a function for  $s, a$ .  $\therefore$  we don't have to consider the probability of  $r$ .  $\therefore V_{\pi}(s) = \sum_a \pi(a|s) \sum_{s'} P(s' | s, a) [r + \gamma V_{\pi}(s')]$

$$\begin{aligned} 3) \quad \bar{V}_{\pi}(s, a) &= E[G_t | S_t = s, A_t = a] \\ &= E[r + \gamma G_{t+1} | S_t = s, A_t = a] \quad (s' \text{ is arbitrary next state of } s) \\ &= E[r + \gamma \sum_{a'} \pi(a' | s') \bar{V}_{\pi}(s', a') | S_t = s, A_t = a] \\ &= \sum_{s', r} P(s', r | s, a) [r + \gamma \sum_{a'} \pi(a' | s') \bar{V}_{\pi}(s', a')] \\ &\quad (\because s', r, a' \text{ are all random variables}) \end{aligned}$$

$$4) \quad q_{\pi}(s, a) = \sum_{s'} p(s'|s, a) [r + \gamma \sum_{a'} \pi(a'|s') q_{\pi}(s', a')]$$

$$5) \quad q_{\pi}(s, a) = \mathbb{E} [r + \gamma G_{t+1} \mid S_t = s, A_t = a],$$

$$G_{t+1} = V_{\pi}(s') \quad (s' \text{ is random variable and next state of } s)$$

$$\therefore q_{\pi}(s, a) = \mathbb{E} [r + \gamma V_{\pi}(s') \mid S_t = s, A_t = a]$$

$$= \sum_{s', r} p(s', r \mid s, a) [r + \gamma V_{\pi}(s')]$$

$$(6) \quad \text{We put } q_{\pi}(s, a) \text{ as } \sum_{s', r} p(s', r \mid s, a) [r + \gamma V_{\pi}(s')] \text{ in (1),}$$

$$\text{Then } V_{\pi}(s) = \sum_a \pi(a \mid s) q_{\pi}(s, a)$$

Problem 3. We should show the Bellman expectation equation must hold which

$$V_{\pi}(s) = \sum_a \pi(a \mid s) \cdot \sum_{s', r} p(s', r \mid s, a) [r + \gamma V_{\pi}(s')],$$

$$\text{Let's consider for 1.9 state. } 1.9 = \frac{1}{4} (\gamma (0.5 + 0.4 + 5.3 + 2.3)) = \frac{\gamma}{4} \times 8.5,$$

$$19 \times 4 = 85 \gamma = 16, \quad \gamma = \frac{16}{85},$$

$$\therefore \frac{1}{4} \left( \frac{16}{85} (0.4 - 0.4 + 2.3 + 0.7) \right) = \frac{19}{85} \times 3.0 = 0.67058 \dots \approx 0.7$$

Problem 4. We can use Bellman equation which

$$q_{\pi}(s, a) = R_s^a + \gamma \sum_{s'} p_{ss'}^a V_{\pi}(s'),$$

$$\therefore q_{\pi}(11, \text{down}) = -1 + 1 \cdot (0.0) = -1,$$

$$q_{\pi}(17, \text{down}) = -1 + 1 \cdot (-14) = -15$$

$$\text{Problem 5. } V_{\pi}(s) = \sum_a \pi(a \mid s) \cdot \left( R_s^a + \gamma \sum_{s'} p_{ss'}^a V_{\pi}(s') \right) \quad (\gamma = 1)$$

$$\therefore V_{\pi}(15) = \frac{1}{4} \left\{ (-1 + V_{\pi}(12)) + (-1 + V_{\pi}(15)) + (-1 + V_{\pi}(14)) + (-1 + V_{\pi}(13)) \right\},$$

$$= -1 + \frac{1}{4} (-22 - 20 - 14) + \frac{1}{4} V_{\pi}(15),$$

$$\frac{3}{4} V_{\pi}(15) = -15, \quad V_{\pi}(15) = -20.$$

$\therefore$  Before adding 15, state 13 has  $V_{\pi}(\text{up}) = -20$ ,  $V_{\pi}(\text{left}) = -22$ ,  $V_{\pi}(\text{right}) = -14$ ,  $V_{\pi}(\text{down}) = -20$ , which is exactly same for state 15 added under 13.  
 $\therefore$  The value function for  $V_{\pi}(15)$  is just same as state 13 which before adding 15.

Problem 6. optimal action  $a$  for state  $s = \arg\max_a \left( R_s^a + \gamma \sum_{s'} P_{ss'}^a V(s') \right)$

But when we add a constant  $C$  to all rewards, then  $R_s^a = -1 + C$  (without  $s = \text{terminate state}$ )

if  $-1 + C < 0$ ,  $C < 1$  then agent find shortest path to maximize cumulative reward.

But if  $-1 + C > 0$ ,  $C > 1$  then agent try to find longest path to maximize cumulative reward.

$\Rightarrow$  for this example the task is totally changed.

$$\begin{aligned} \text{Problem 7. } \bar{J}_{k+1}(s, a) &= R_s^a + \gamma \sum_{s'} P_{ss'}^a V_{k+1}(s') \\ &= \sum_{s', r} P(s', r | s, a) (r + \gamma V_{k+1}(s')), \end{aligned}$$

$$\therefore \sum_{s', r} P(s', r | s, a) [r + \gamma V_k(s')] = \bar{J}_k(s, a)$$

$$\therefore V_{k+1}(s) = \max_a \bar{J}_k(s, a)$$

$$\therefore \bar{J}_{k+1}(s, a) = \sum_{s', r} P(s', r | s, a) \left[ r + \gamma \max_{a'} \bar{J}_k(s', a') \right]$$

$$\text{Problem 8. } \bar{J}_{\pi}(s, a) = E_{\pi} [G_t | S_t = s, A_t = a]$$

$$= E_{\pi} [R_{t+1} + \gamma G_{t+1} | S_t = s, A_t = a]$$

$$= E_{\pi} [R_{t+1} + \gamma \bar{J}_{\pi}(S_{t+1}, A_{t+1}) | S_t = s, A_t = a],$$

$$\bar{J}_{\pi}(s, a) = \sum_{s', r} P(s', r | s, a) [r + \gamma \bar{J}_{\pi}(s', A_{t+1})]$$

$$= \sum_{s', r} P(s', r | s, a) \left[ r + \gamma \sum_{a'} \pi(a' | s') \bar{J}_{\pi}(s', a') \right]$$

$$\therefore \bar{J}_{k+1}(s, a) = \sum_{s', r} P(s', r | s, a) \left[ r + \gamma \sum_{a'} \pi(a' | s') \bar{J}_k(s', a') \right]$$

Problem 9. let the state which should make decision :  $S$ , let the state after  $S$  when choosing left:  $S_L$ , and let the state after  $S$  when choosing right:  $S_R$ .

$$\text{if we follow } \pi_{\text{left}}, V(s) = 1 + r \cdot (0 + r(1 + r(0 + \dots)) \\ = 1 + r^2 + r^4 + \dots = \sum_{n=0}^{\infty} r^{2n} = \frac{1}{1-r^2} \quad (\because r < 1)$$

$$\text{if we follow } \pi_{\text{right}}, V(s) = 0 + r \cdot (2 + r(0 + r(2 + \dots)) = \\ = 2r + 2r^3 + 2r^5 + \dots = \sum_{n=0}^{\infty} 2r^{2n+1} = \frac{2r}{1-r^2} \quad (\because r < 1)$$

$$\frac{2r}{1-r^2} > \frac{1}{1-r^2} \Rightarrow 2r > 1, \quad r > \frac{1}{2} \quad (\because 1-r^2 > 0)$$

$\therefore$  if  $r=0$  then  $\pi_{\text{left}}$  is optimal,

$r=0.9$  then  $\pi_{\text{right}}$  is optimal,

$r=0.5$  then Both  $\pi_{\text{left}}, \pi_{\text{right}}$  are optimal.